Engineering Notes - Part 1: Python Essentials

A virtual environment isolates the packages of one project from the rest of the machine, so different projects can use different library versions without conflicts. Create it with python -m venv venv and activate it before installing packages with pip.

A list comprehension builds a new list in a single expression, for example squares = [n * n for n in numbers if n > 0]. It is shorter than a loop and usually easier to read for simple transformations.

Dictionaries map keys to values and give fast lookups. Use the get method to read a key that may be missing, and items to loop over keys and values together.

Exceptions are handled with try and except blocks. Catch the specific error you expect, such as ValueError or KeyError, instead of catching everything, so real bugs are not hidden.

A decorator is a function that wraps another function to add behavior such as logging, timing, or caching. FastAPI uses decorators like app.get to register endpoints.

Engineering Notes - Part 2: SQL Basics

SELECT chooses the columns to return and FROM names the table. WHERE filters rows before grouping, and ORDER BY sorts the result.

A JOIN combines rows from two tables using a related column. An inner join returns only the rows that have a match in both tables, while a left join keeps every row from the left table even when there is no match.

GROUP BY collects rows that share a value so that aggregate functions such as COUNT, SUM, AVG, MIN, and MAX can be computed per group. HAVING filters groups after aggregation, while WHERE filters individual rows before it.

A primary key uniquely identifies each row in a table. A foreign key refers to the primary key of another table and keeps the data consistent. An index speeds up lookups on a column but makes writes slightly slower and uses extra storage.

Engineering Notes - Part 3: Web APIs

A GET request reads data and puts its parameters in the URL, while a POST request sends data in the request body and is used to create or process something. Status code 200 means success, 404 means the resource was not found, 422 means the request body failed validation, and 500 means an unexpected server error.

Pydantic validates incoming data against a model. If a required field is missing or has the wrong type, FastAPI rejects the request automatically with a 422 response before your function runs.

APIs that call paid services should use authentication with an API key and rate limiting, otherwise anyone who finds the URL can spend your money. Keys should be sent in a header and never in the URL.

A health check endpoint returns a simple status so that a platform can verify the service is running. Logging every error with enough context makes problems easier to debug in production.

Engineering Notes - Part 4: Docker and Deployment

A Docker image is a read-only template that contains the code, the runtime, and the dependencies. A container is a running instance of an image. You can start many containers from the same image.

A Dockerfile lists the steps to build an image. Copy the requirements file and install dependencies before copying the application code so Docker can reuse its cache when only the code changes.

Secrets such as passwords and API keys must be passed as environment variables and never committed to the repository or written inside the image. Add the environment file to gitignore before the first commit.

Continuous integration runs the tests automatically on every push, and continuous delivery deploys the application when the tests pass. Pin the dependency versions so builds are reproducible.